

Advanced Programming Practices

MTech 2026

Lab - 3

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP3_Qn.c (e.g. ABC2026001_APP3_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. A magic square is a square grid of numbers where the sum of the numbers in each row, column, and diagonal is the same. Write a C program to implement the magic square for the $n \times n$ matrix, where n is an odd number and the array contains numbers 1 to n^2 .
2. Write a C program to swap two numbers. Make two functions named as `call_by_value` and `call_by_reference` to pass the value/reference of two variables for swapping.
3. Write a C program to print a string in reverse order using recursion. You need not store the characters of the string in an array or any other data structure.
4. Write a C program to calculate the length of the input string using recursion.
5. Write a C program that takes a string as input and compress the consecutive characters in the input string. Example: input: aaabbcddddd output: a3b2c1d5a1.